

GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

SEMESTER V

Name of Department: - Computer Science and Engineering

Course Code:		TCS 504		Course Title:		System Design		
Credits:	3	Semester:		V	Category of Course:		DSC	
TEACHING PERIODS (In hours per week)				TW+SL (in hours per semester)		Relative Weight		
L	T	P		45		CIE	MSE	ESE
3	0	0				25	25	50
Examination Duration (Hrs):				Theory			Practical	
				3			0	

Pre-requisite: DBMS (TCS 503), Operating Systems (TCS 502), Computer Networks (TCS 514).

Course Outcome:	After completion of the course, the students will be able to: CO1: Analyze the scalability, database consistency, and communication protocols CO2: Design efficient and secure data handling and asynchronous messaging workflows using NoSQL databases and message queues CO3: Evaluate architectural patterns, rate-limiting algorithms, and distributed caching. CO4: Implement microservice decomposition patterns, service discovery mechanisms, and fault-tolerance strategies CO5: Synthesize large-scale architectures, formulating technical schemas and infrastructure components for real world services or applications
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Details of the Course:

Sl. No.	Contents	Contact Hours
1	Unit 1: The Foundations of Scalability Architecture Basics: Evolution from Monolithic to N-Tier Architecture; Vertical Scaling (Scale-up), Horizontal Scaling (Scale-out); Latency, Throughput (Disk seek, L1 cache ref, network trip). Networking & Communication: Communication Protocols-REST, RPC, GraphQL, WebSockets; Load Balancing - L4 (Transport Layer), L7 (Application Layer); Algorithms- Round Robin, Least Connections, Consistent Hashing.	10

	Caching Strategies: Reading Strategies- Cache-Aside, Read-Through; Writing Strategies- Write-Through, Write-Back, Write-Around; Eviction Policies- LRU, LFU; Tool Focus- Redis/Memcached architecture basics. Database Concepts for Scale: ACID (SQL), BASE (NoSQL) consistency models; CAP Theorem- Understanding Trade-offs (CP vs AP systems); Replication- Master-Slave, Master-Master; Sharding-Horizontal partitioning, Sharding keys, and Hotspot problems.	
2	Unit 2:Data Handling & Asynchronous Processing NoSQL Deep Dive: Types of NoSQL databases and their specific use cases-Key-Value, Document, Columnar, Graph. Asynchronous Messaging & Queues: Synchronous and Asynchronous processing; Message Queues- Decoupling producers and consumers (RabbitMQ), Pub/Sub Models- Fan-out architecture (Kafka).; Handling backpressure and dead-letter queues. Identity & Security: Authentication, Authorizatio; Session-based Auth, Token-based Auth (JWT); OAuth 2.0, OpenID Connect flows; HTTPS/SSL Handshake basics in a distributed context.	9
3	Unit 3: Case Studies Case Study: Design a URL Shortener: Key Challenges- Generating unique IDs, handling heavy read-heavy traffic; Application- Using KGS (Key Generation Service) to pre-generate keys; Bottleneck Analysis-Cache eviction policies for "hot" URLs. Case Study: Design a Rate Limiter: Key Challenges- Preventing DDOS and API abuse; Algorithms- Token Bucket, Leaky Bucket, Sliding Window; Application- Using Redis for distributed counting. Lab/Workshop: Schema Design: Activity- Students must design the Database Schema (SQL vs NoSQL) for a "Pastebin" service and justify their choice based on Unit 1 concepts (CAP theorem).	8
4	Unit 4: Advanced Reliability & Microservices Microservices Architecture: Monolith services and Micro services - Decomposition patterns; Service Discovery- Client-side discovery, Server-side discovery.API Gateway- Rate limiting, routing, and aggregation at the edge. Distributed Transactions & Consistency: Problem of distributed commits; Two-Phase Commit - Risks and blocking; Saga Pattern- Choreography, Orchestration for long-running transactions. Reliability & Observability: Resiliency Patterns- Circuit Breaker, Bulkhead, Retry with Exponential Backoff; Observability- Distributed Tracing, Logging, and Metrics. Content Delivery & Storage: Content Delivery Networks – Push, Pull CDNs; Blob Storage - Storing large media files (S3 structure, multipart uploads).	10
5	Unit 5: Advanced Case Studies Case Study: Design a Social Media Feed (Instagram/Twitter): Key Challenges- Feed generation (Pull vs. Push/Fan-out on write); Application- Hybrid approach for celebrities and normal users; Storage- Storing graph data (Followers) vs. Blob data (Images). Case Study: Design a Ride-Sharing Service (Uber/Lyft): Key Challenges- Real-time location updates, Geospatial databases (QuadTrees vs. Geohashes); Application: Matching drivers to riders efficiently; Consistency - Handling "Double Booking" using distributed locks; Case Study: Design a Chat Application (WhatsApp/Discord): Key Challenges- Real-time delivery, presence indicators (Online/Offline); Application- WebSocket connections, preserving message order, historical chat storage (Cassandra/HBase).	8
	Total	45

Text Books:

Authors Name	Title	Edition	Publisher, Country	Year
Alex Xu	System Design Interview: An Insider's Guide	Second	Independently published	
Alex Xu	System Design Interview: An Insider's Guide Volume 2	Second	Independently published	